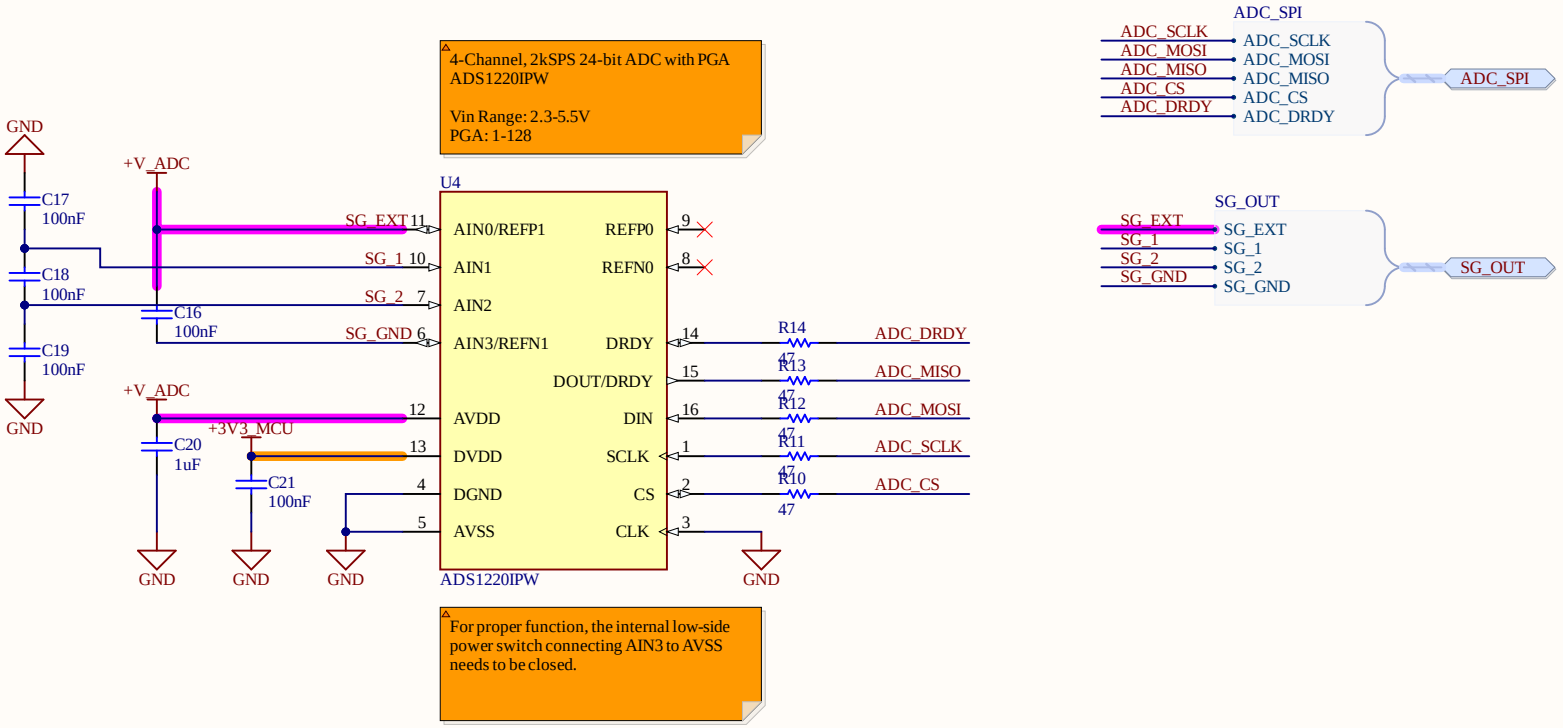
FOLIA CTRL BOARD
ADC for Strain Gauge



Title	adc.SchDoc		
Size: Letter	Number:	Revision: 4	
Date: 9/29/2025	Time: 5:44:02 PM	Author: D. Lee, A. Ladd	
File: adc.SchDoc			

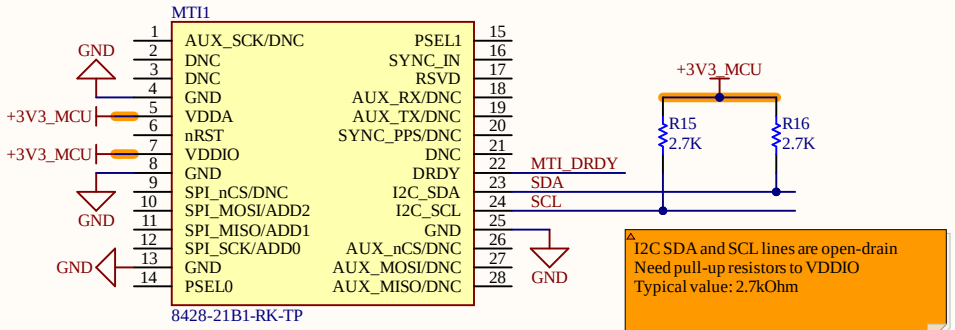
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75 Amherst Street
Cambridge, MA 02139
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FOLIA CTRL BOARD
IMU

MTI-1 Series IMU (Xsens by Movella)

Vin Range: 3.3V (2.8-3.6V)
Protocols: I2C, SPI, UART



I2C SDA and SCL lines are open-drain
Need pull-up resistors to VDDIO
Typical value: 2.7kOhm



PSEL Connections:
I2C: PSEL0 & PSEL1 DNC
SPI: PSEL0 GND, PSEL1 DNC
UART half-duplex: PSEL0 DNC, PSEL1 GND
UART full-duplex: PSEL0 & PSEL1 GND

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FOLIA CTRL BOARD
Arduino Portenta H7

80-pin High Density Headers

ADC_SPI
ADC_SCLK
ADC_MOSI
ADC_MISO
ADC_CS
ADC_DRDY
ADC_SPI

EXT_UART_MOT
UART_MOT_TX
UART_MOT_RX
EXT_UART_MOT

UI
LED1
LED2
UI

MTI_I2C
IMU_SDA
IMU_SCL
IMU_DRDY
MTI_I2C

EXT_PINS
I2C2_SCL
I2C2_SDA
TX2
RX2
GPIO3
GPIO4
PWM2
PWM3
EXT_PINS

P1

2	2	ETH_A+	1
4	4	ETH_A-	3
6	6	ETH_B+	5
8	8	ETH_B-	7
10	DSI_D1+	9	11
12	DSI_D1-	11	13
14	DSI_D0+	13	15
16	DSI_D0-	15	17
18	DSI_CK+	ETH_L1	19
20	DSI_CK-	ETH_L2	21
22	GND	VIN	23
24	24	USB1_VBUS	25
26	USB0_D+	USB1_D+	27
28	USB0_D-	USB1_D-	29
30	30	USB1_ID	31
32	VIN	GND	33
34	UART0_TX	UART1_TX	35
36	UART0_RX	UART1_RX	37
38	UART0_RTS	UART1_RTS	39
40	UART0_CTS	UART1_CTS	41
42	GND	VIN	43
44	I2C0_SDA	I2C1_SDA	45
46	I2C0_SCL	I2C1_SCL	47
48	VIN	GND	49
50	50	CAN1_TX	51
52	52	CAN1_RX	53
54	GND	VSYS	55
56	I2S_MCK	SDC_CLK	57
58	I2S_WS	SDC_CMD	59
60	I2S_DI	SDC_D0	61
62	I2S_DO	SDC_D1	63
64	VSYS	SDC_D2	65
66	DMIC_CK	SDC_D3	67
68	DMIC_D0	67	69
70	70	69	71
72	V-SDCARD	71	73
74	74	RESET	75
76	76	SWDIO	77
78	78	SWCK	79
80	80	SWO	

DF40C-80DS-0.4V(51)

P2

79	ADC_A3	ADC_A7	80
77	ADC_A2	ADC_A6	78
75	ADC_A1	ADC_A5	76
73	ADC_A0	ADC_A4	74
71	ADC_VREF+	ADC_VREF-	72
69	VCC/+3V3	GND	70
67	PWM_4	PWM_9	68
65	PWM_3	PWM_8	66
63	PWM_2	PWM_7	64
61	PWM_1	PWM_6	62
59	PWM_0	PWM_5	60
57	GND	GPIO_6	58
55	55	GPIO_5	56
53	SAI_D0	GPIO_4	54 GPIO4
51	SAI_FS	GPIO_3	52 GPIO3
49	SAI_CK	GPIO_2	50 IMU_DRDY
47	I2C2_SCL	GPIO_1	48
45	I2C2_SDA	GPIO_0	46 ADC_DRDY
43	VCC/+3V3	GND	44
41	41	SPI1_COPI	42 ADC_MOSI
39	39	SPI1_CIPO	40 ADC_MISO
37	37	SPI1_CS	38 ADC_SCLK
35	35	SPI1_CS	36 ADC_CS
33	GND	VCC	34
31	31	32	32
29	29	30	30
27	UART3_RX	UART2_RX	28 RX2
25	UART3_TX	UART2_TX	26 TX2
23	VCC/+3V3	GND	24
21	21	CAM_HS	22
19	19	CAM_CLK	20
17	17	CAM_VS	18
15	15	CAM_D0	16
13	13	CAM_D1	14
11	11	CAM_D2	12
9	9	CAM_D3	10
7	COINCELL	CAM_D4	8
5	PWR_ON_REQ	CAM_D5	6
3	BOOT_SRC	CAM_D6	4
1	FORCE_BTLDR	CAM_D7	2

DF40C-80DS-0.4V(51)

2.54mm Header pins

P3

1	1	+5V_USB
2	2	+5V_PWR
3	3	
4	4	GND
5	5	
6	6	
7	7	
8	8	
9	9	
10	10	
11	11	
12	12	
13	13	
14	14	

22-28-4143

P4

1	1
2	2
3	3
4	4
5	5
6	6
7	7
8	8
9	9
10	10
11	11
12	12
13	13
14	14

22-28-4143

Arduino Portenta H7
Connections

Title	mcu.SchDoc		
Size:	Letter	Number:	Revision:4
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FOLIA CTRL BOARD
POWER

Arrows indicate general direction of power flow

Battery input

J1
XT30PW-M
P
N
+VB

C1 10uF
C2 220uF
C3 10uF
C4 100nF
GND

J2
XT30PW-F
P
N

Motor output

C3, C4 physically next to U1 (buck) Vin pin

+VB
C5 100nF
VIN
CB
SW
SHDN
GND
FB
L1
SRR0805-5R6M
+5V_PWR
R3 88.7K
R4 22.1K
C6 10uF
C7 10nF
S1
U1
LMR50410YQDBVRQ1
Wide-Range SMPS Buck

Wide-Range SMPS
LMR50410YFQDBVRQ1

Vin Range : 4V-36V
Vin Net : (+VB)
Vout : 5V (+5V_PWR)
Cin: 2.2 uF
Cout: 10 nF
Cboot: 100 nF

Imax : 1 A

If the +5V_PWR supply is available, it powers the circuit. Otherwise, +5V_USB is used.

ST is pulled HIGH when VIN1 is being used and LOW when VIN2 is being used.

+5V_PWR
C8 1uF
R5 22.1K
+5V_USB
C9 1uF
R6 5.76k
Autoswitch Power Mux
U2
VIN1
VIN2
PR1
MODE
VOUT
VOUT
ST
GND
TPS2116DRLR
+5V
C10 100nF
C11 10uF

Auto-switch Power Mux
TPS2116DRLR

Vin Range : 1.6-5.5V
Vin Net : +5V_PWR or +5V_USB
Vout : 5V (+5V)
Cin: 1 uF
Cout: 10 uF

Imax : 2.5A

Automatic priority mode prioritizes the supply connected to VIN1 and switches over to the secondary supply (VIN2) when VIN1 drops. When PR1 is pulled above (VREF = 1), the voltage on VIN1 is used to power the output, and when it is pulled below VREF, VIN2 is used to power the output.

With 22.1k & 5.76k it will switch around 4.8V.

Cout: 10 uF
Cin: 1 uF

+5V
ADC LDO
U3
+VI
+VO
GND
FB
EN
BYPASS
TPS79301DBVRQ1
C12 1uF
C13 10nF
R7 43.7k
R8 30.1k
C14 16pF
C15 1uF
+V_ADC

LDO
TPS79301DBVRQ1

Vin Range : 2.7V-5.5V
Vin Net : (+5V)
Vout : 1.2-5.5V (+V_ADC)
Vref: 1.225V Typ

Imax : 200mA
112mV dropout (@500mA)
32uVrms noise
PSRR : 700db (@10khz)

Supplying ADC power + St.Gauge Excitation

If R8=43.7k and R9=30.1k
Vo = Vref *(1+R8/R9) = 3.00V
C14 = (3*10^(-7) * (R8+R9)) / (R26*R27)
C14 = 16.8pF

Power indicator diode

+V_ADC
R9 3k
D3
APT1608EC
GND

EXT_PWR
+5V_PWR
I2C2_SDA
+V_ADC
+5V_PWR
+3V3_MCU
+V_ADC
EXT_PWR

Title	pwr.SchDoc		
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FOLIA CTRL BOARD
TOP

U_MCU
mcu.SchDoc

MTI_I2C

U_IMU
imu.SchDoc
MTI_I2C

ADC_SPI

U_ADC
adc.SchDoc
ADC_SPI
SG_OUT
SG_OUT
SG_EXT
SG_1
SG_2
SG_GND
SG_EXT
SG_1
SG_2
SG_GND

EXT_UART_MOT

EXT_UART_MOT
UART_MOT_TX
UART_MOT_RX
UART_MOT_TX
UART_MOT_RX

EXT_PINS

EXT_PINS
I2C2_SCL
I2C2_SDA
TX2
RX2
GPIO3
GPIO4
PWM2
PWM3
I2C2_SCL
I2C2_SDA
TX2
RX2
GPIO3
GPIO4
PWM2
PWM3

UI

UI
LED1
LED2
LED1
LED2

U_PWR
pwr.SchDoc

EXT_PWR

EXT_PWR
+5V_PWR
+3V3_MCU
+V_ADC
+5V_PWR
+3V3_MCU
+V_ADC

2.54mm Headers

P5
+5V_PWR 1
+V_ADC 3
RX2 5
GPIO4 7
PWM2 9
I2C2_SDA 11
2 +3V3_MCU
4 GND
6 TX2
8 GPIO3
10 PWM3
12 I2C2_SCL
61301221121

Strain Gauge Molex Connector

J3
SG_GND 1
SG_2 2
SG_1 3
SG_EXT 4
1
2
3
4
S1
S2
SHIELD
SHIELD
0532610471

Motor Molex Connector

+3V3_MCU
J4
UART_MOT_TX 2
UART_MOT_RX 3
1
2
3
4
GND
S1
S2
SHIELD
SHIELD
0532610471

UI LEDs

LED1
R1
1k
D1
GND
APT1608SGC

LED2
R2
1k
D2
GND
APT1608SGC

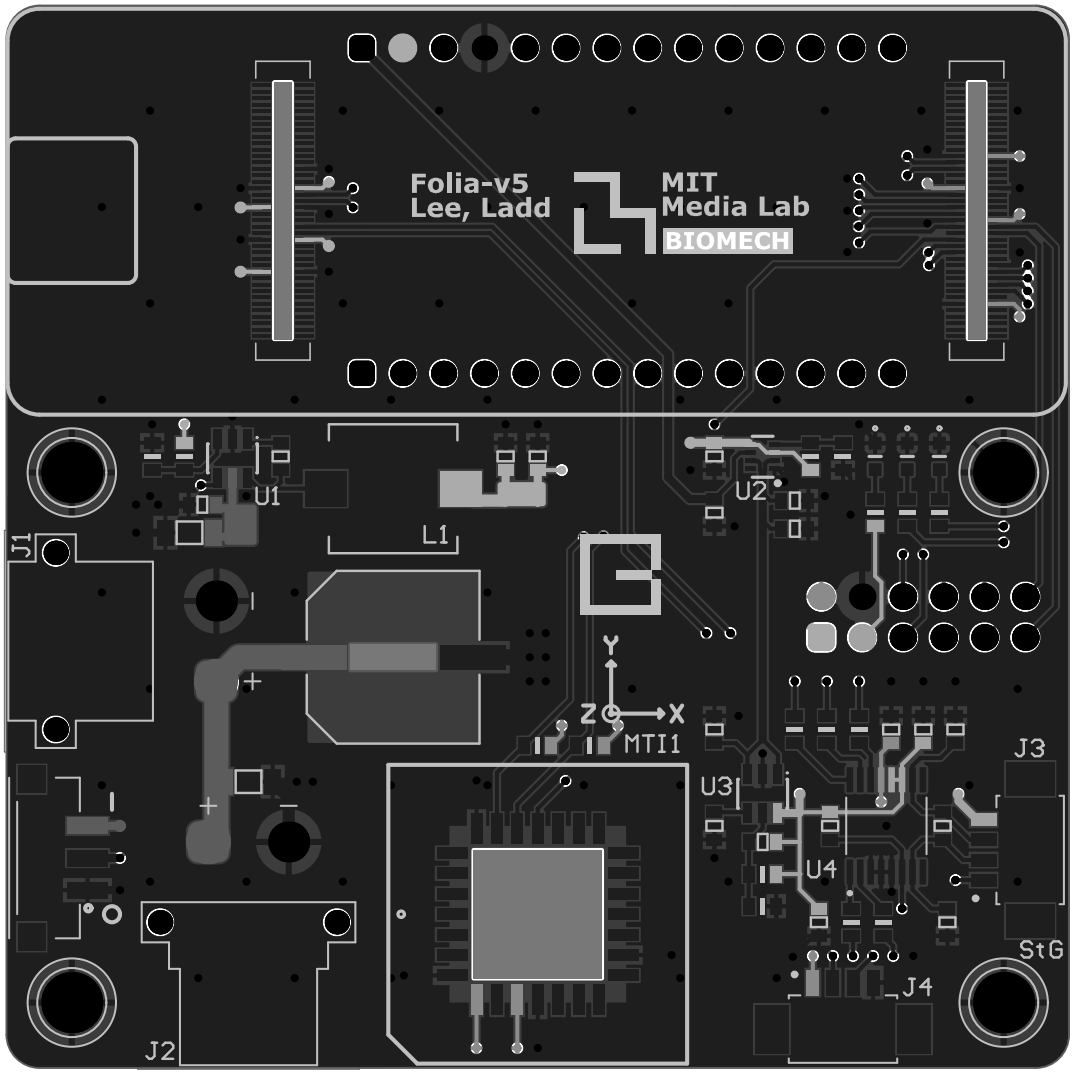
Mounting Holes

GND
M1
GND
M3
GND
M2
GND
M4

Title	top.SchDoc		
Size: Letter	Number:	Revision:4	
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Folia-v5
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MIT
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BIOMECH
U1
L1
U2
J1
Y
Z
X
MTI1
U3
J3
U4
J4
StG
J2